DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

Site Analysis & Human-Centric Research

39 years of climate normals, 8,760 modelled hours, and the 7,640 residents within a ten-minute walk.

15,000 m²	44.5% → 64.6%	−7.13 °C	131
SITE AREA	COMFORTABLE DAYLIGHT HOURS	HEAT INDEX UNDER CANOPY	TREES PLANTED

▶ VISIT THE LIVE PROJECT PORTAL wasim437.github.io/AI_SAFA/

01 WHAT IS IN THIS FILE

Phase2 Problem Definition Report	DRAWING
Fig01 climate and comfort	FIGURE
Fig09 diurnal comfort	FIGURE

02 THE SCALE OF THE ANALYSIS BEHIND IT

39 years
of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours
of sun position ray-traced against 131 trees for every hour of the year

15,000
one-metre ground cells modelled for shade and comfort, site-wide

4 models
trained and tested for leakage — two of them decide what this document reports

41 checks
run on every rebuild, so a wrong number fails loudly instead of shipping quietly

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/AI_SAFA
Live portal	wasim437.github.io/AI_SAFA/
Produced by	CODE src/climate.py src/solar.py DATA data/raw/sources.json DOCS DATA_SOURCES.md
Reproduce	`python run_analysis.py` · `python -m tests.test_pipeline`

Generated 2026-08-03 · reproducible with python tools/build_submission_pdfs.py

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

01 THE TEN PHASES

P1 Site & Context Analysis — THIS DOCUMENT
39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

P2 Problem Definition
Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

P3 Opportunity & Objectives
The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

P4 Concept Development
Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

P5 Masterplan Development
Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

P6 Detailed Design
The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

P7 Performance & Sustainability
Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

P8 User Experience & Activation
K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

P9 AI Workflow & Visualisation
The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

P10 Submission Assembly
Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02 THE CODE AND DATA BEHIND THIS DOCUMENT

- **Climate** — *src*
- **Solar** — *src*
- **Sources** — *data/raw*
- **DATA SOURCES** — *DATA_SOURCES.md*

03 REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 11 OF 12 · SITE ANALYSIS & HUMAN-CENTRIC RESEARCH

Site Analysis & Human-Centric Research

What the site is, who it serves, and what is wrong with it

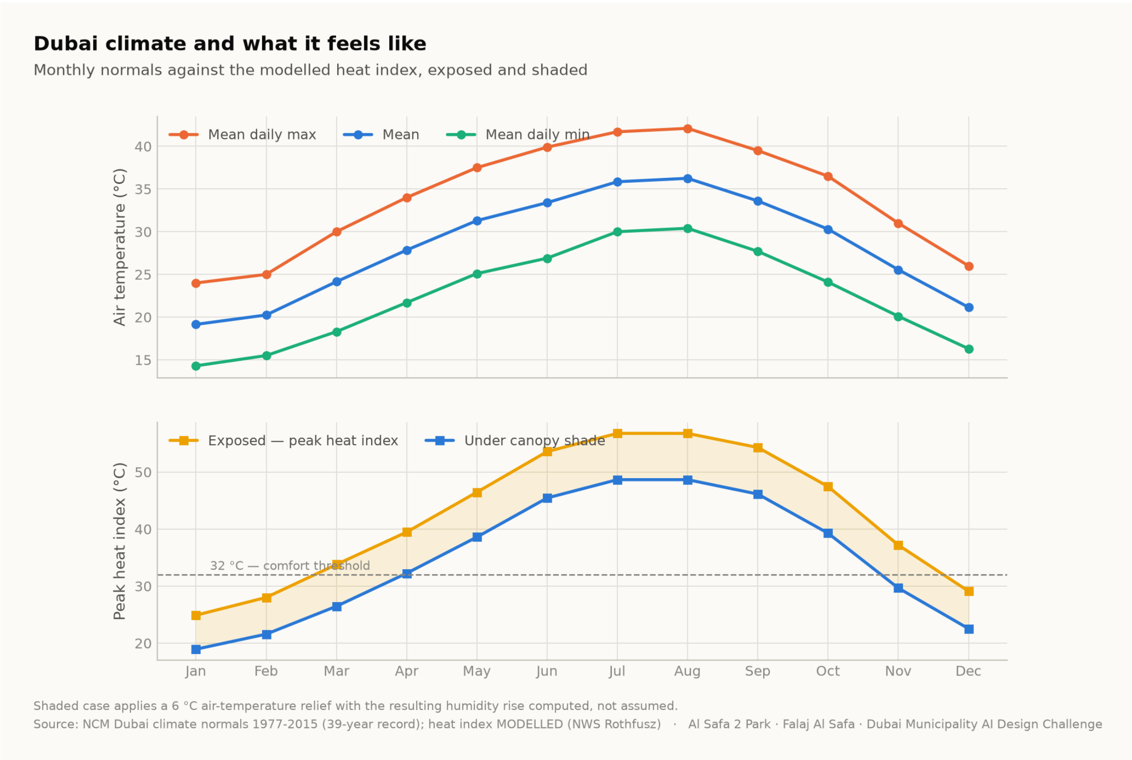
> A flat, largely exposed 15,000 m² neighbourhood park serving roughly 7,640 residents within a ten-minute walk. Its defining problem is not layout or planting: for 55.5% of daylight hours, standing outside on it is uncomfortable or worse.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 03 August 2026 · every quantity regenerated by `python run_analysis.py`

1. Climate — the governing constraint

The analysis reconstructs an 8,760-hour year for the site from 39 years of National Centre of Meteorology monthly normals, with solar positions computed for every hour by the NREL algorithm at the site's coordinates. The reconstruction is verified back against the published normals to within 0.39 °C.

The climate series is modelled, not measured. The underlying record is 39 years long but is published as monthly normals; the hourly series is reconstructed from them and labelled as modelled everywhere it appears. Conclusions are about a typical year, never about extremes.



Dubai climate normals against the modelled heat index. Analysis output — computed from project data.

2. The problem, ranked rather than listed

Problem	Severity	Evidence
Thermal discomfort	Critical	Only 44.5% of 4,402 daylight hours are comfortable; peak heat index 56.8 °C
No continuous shaded route	Critical	A straight or fragmented shade strategy leaves 330 hours a year with no shade anywhere along the primary route
Exposed perimeter	High	Roads on the boundary contribute noise, glare and reflected heat with no intervening mass
Water and irrigation demand	High	Open water or irrigated turf in this climate carries a continuing evaporation and supply cost
Limited programme range	Moderate	Without comfort, no amount of programming produces use

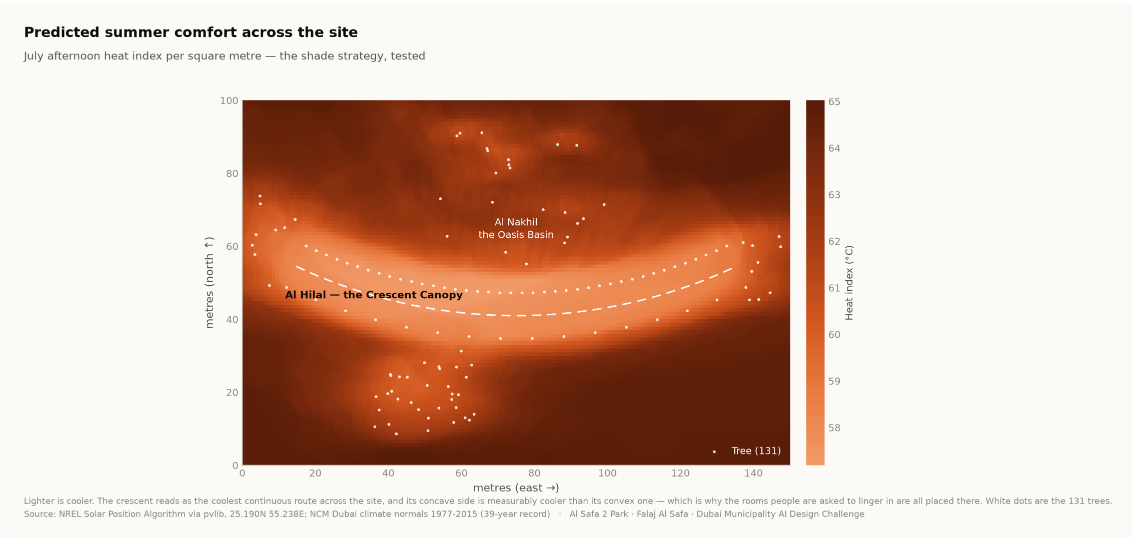
Ranking by severity is what justifies making shade the organising idea rather than one strategy among several.

3. Who the park serves

Approximately 7,640 residents live within a ten-minute walk (Dubai Statistics Center, 2023). The neighbourhood is established and low-rise, with families, working adults, domestic staff and a significant older population. The user groups and their needs are set out in the User Experience and Activation Strategy.

Visitor demand is a scenario, not a prediction. No footfall data exists for this site. Demand figures are deliberately excluded from the machine learning suite, because a model trained on an assumed demand curve would only recover the assumption that produced it.

4. Where the site is comfortable, square metre by square metre



Predicted July afternoon heat index per m². The crescent reads as the coolest continuous route, and its concave side is measurably cooler than its convex face. Analysis output — computed from project data.

5. Opportunities the analysis identifies

- **One continuous shaded route is worth more than dispersed shade.** Permutation importance puts distance to the crescent far above every other geometric feature.
- **Comfort is predictable from the calendar.** At 97.5% accuracy from sun position and date alone, park operations need no sensor network.
- **The shoulder seasons are recoverable.** The comfort gain concentrates in late afternoon in spring and autumn — hours that are currently lost and are cheap to win back.
- **The perimeter is an asset.** A planted berm addresses noise, glare and heat simultaneously.

Fig01 climate and comfort

Analysis output — computed from project data.

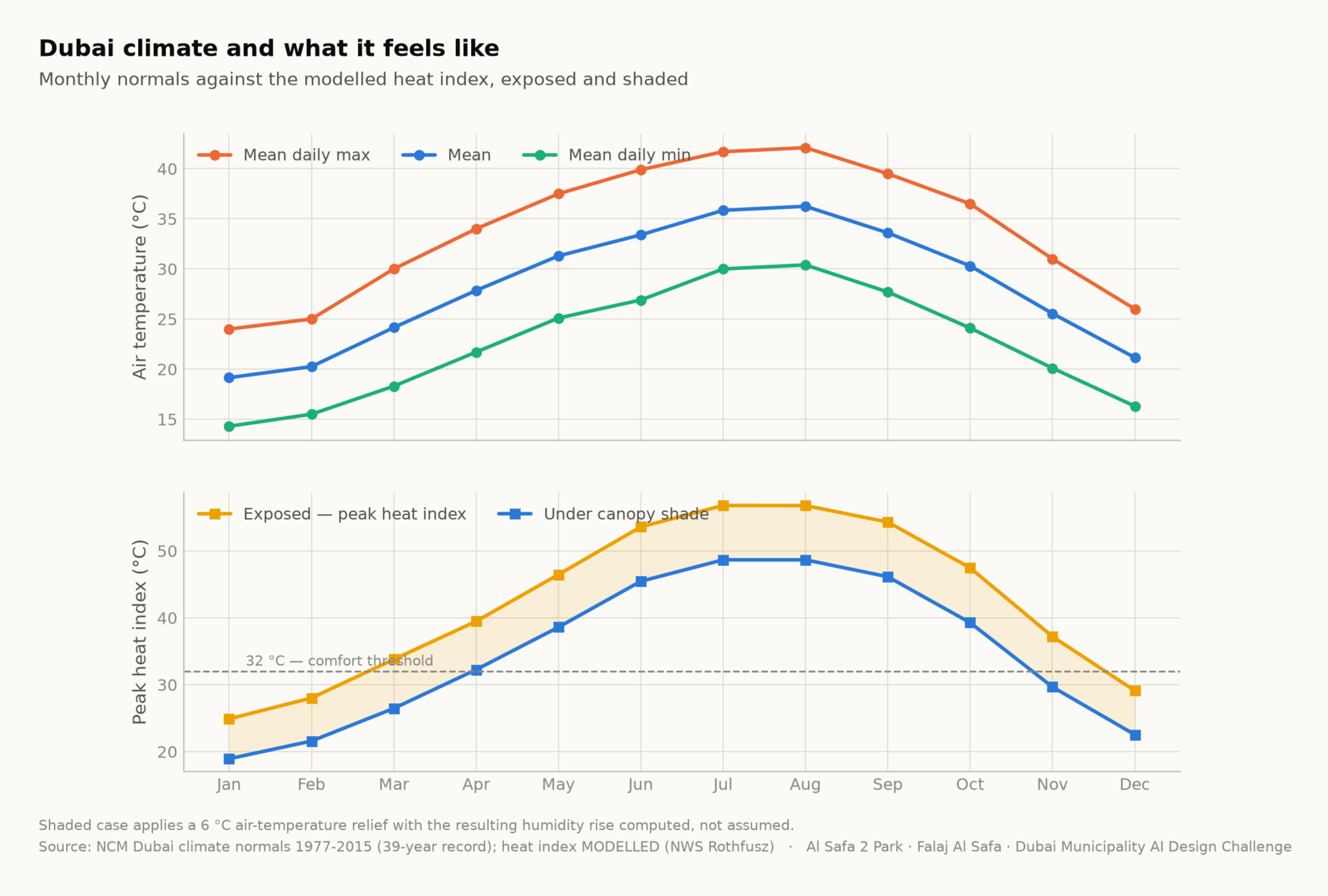
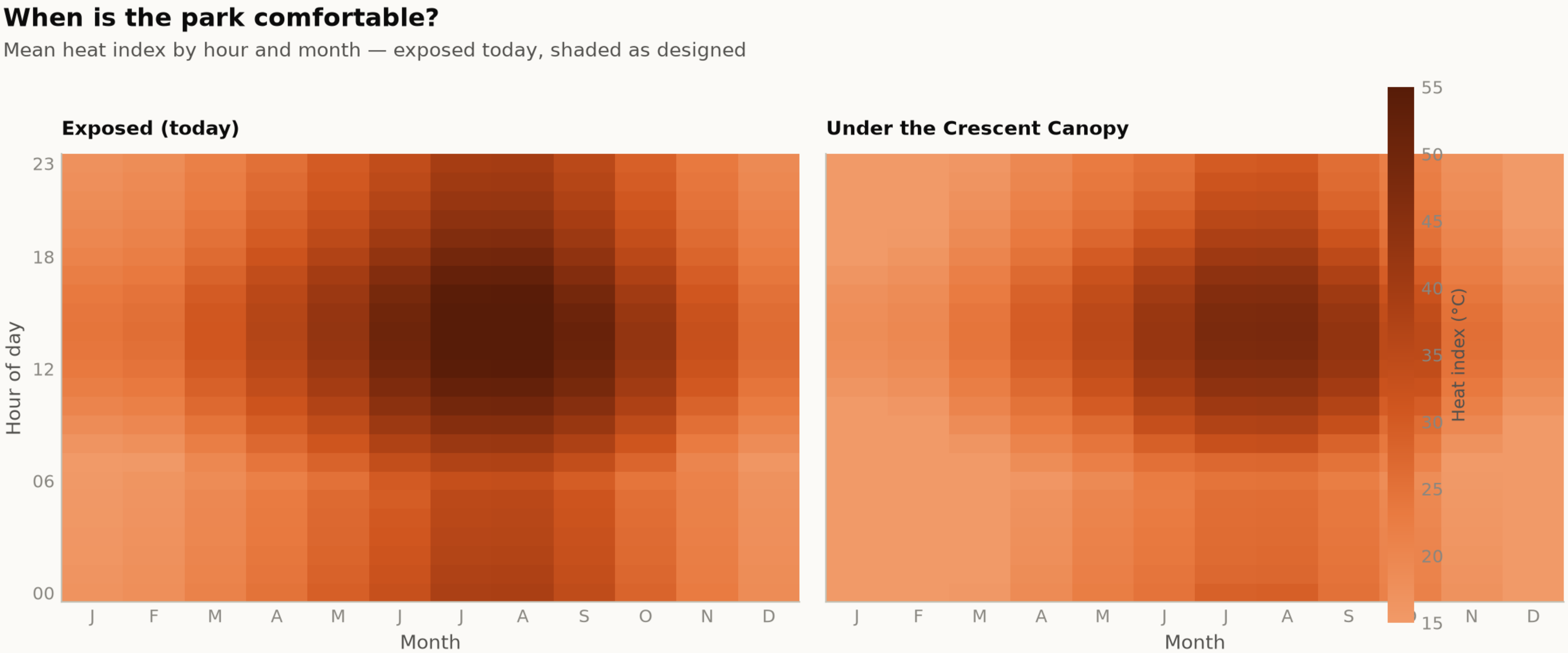


Fig09 diurnal comfort

Analysis output — computed from project data.



The design opens up the late afternoon in spring and autumn, which is when the catchment most wants to use the park.

Source: NCM Dubai climate normals 1977-2015 (39-year record) — hourly series MODELLED, see src/climate.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Every drawing and every chart in this project

All 18 images the pipeline produces, whichever slot they belong to — each labelled with its class and linked to the full-resolution original. Nothing here is hand-drawn.

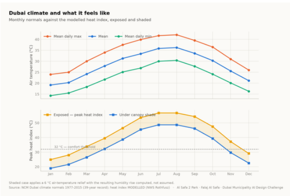


Fig01 climate and comfort
analysis output

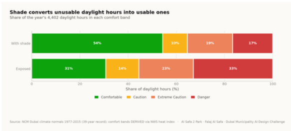


Fig02 comfort bands
analysis output

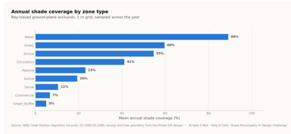


Fig03 shade by zone
analysis output

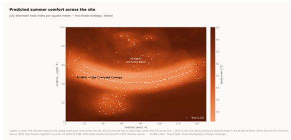


Fig04 site comfort map
analysis output

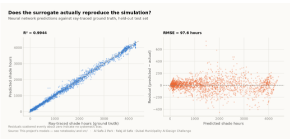


Fig05 surrogate performanc
analysis output

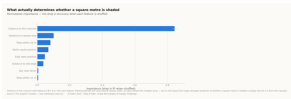


Fig06 feature importance
analysis output

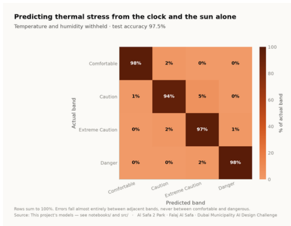


Fig07 confusion matrix
analysis output

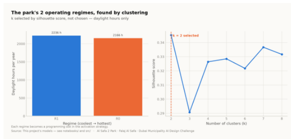


Fig08 microclimate regimes
analysis output

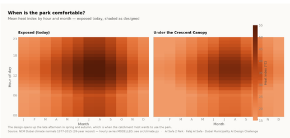


Fig09 diurnal comfort
analysis output

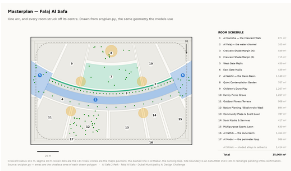


Fig10 masterplan
analysis output

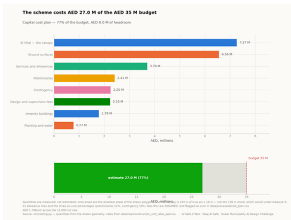
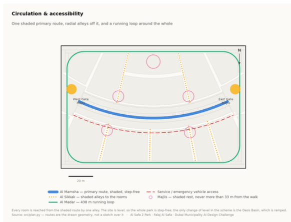
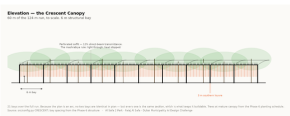


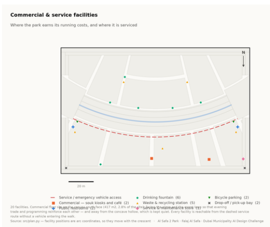
Fig11 cost plan
analysis output



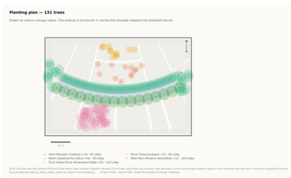
Circulation
technical drawing



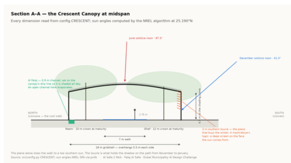
Elevation
technical drawing



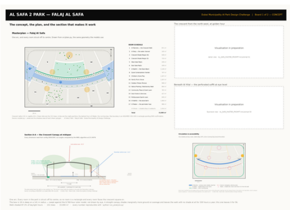
Facilities
technical drawing



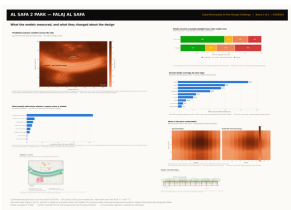
Planting
technical drawing



Section
technical drawing



Board 1 concept
presentation board



Board 2 evidence
presentation board

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

MEASURED RESULT

models/headline_metrics.json

Annual daylight hours modelled	**4,402**
Comfortable daylight hours — today	**44.5%**
Comfortable daylight hours — as designed	**64.6%**
Mean heat-index reduction under canopy	**7.13 °C**
Peak heat index, exposed	**56.8 °C**
Peak heat index, shaded	**48.7 °C**
Crescent Walk shaded, canopy and louvre	**87.3%**
Site-wide mean shade	**34.1%**
Trees planted	**131**

TRAINED MODELS, ON HELD-OUT TEST SETS

models/model_metrics.json

M1a Random Forest — shade surrogate, test R^2	**0.9982**
M1b Neural network — deployed surrogate, test R^2	**0.9944**
M2 Gradient Boosting — comfort band, test accuracy	**97.5%**
M3 K-Means — regimes selected by silhouette	**k = 2**
Training / validation / test split	**10,500 / 2,250 / 2,250**
Random seed, fixed for reproducibility	**42**

WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

Climate fidelity
the modelled year reproduces the published normals it was built from

No overlaps
no room in the schedule overlaps another

Areas close
every area sums back to 15,000 m²

Budget held
the cost plan stays inside AED 35 M

No leakage
no model can see its own answer in its inputs

RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js    # 64 portal checks
node tests/test_film.js          # every frame of the film
```

The complete project, and where to check it

Every one of the 120 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository	**github.com/wasim437/Al_SAFA**
Live portal	**wasim437.github.io/Al_SAFA/**

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Ma…
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Di…
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (7)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/Al_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

figures/fig11_cost_plan.png	Analysis output — computed from project data

THE TECHNICAL DRAWINGS AND BOARDS (7)

design/visuals/circulation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/elevation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/facilities_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/planting_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/section_crescent.png	Technical drawing — generated from src/plan.py
design/boards/board_1_concept.png	Technical drawing — generated from src/plan.py
design/boards/board_2_evidence.png	Technical drawing — generated from src/plan.py

THE ANALYSIS CODE (13)

src/__init__.py	Analysis module
src/boards.py	The two presentation boards
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/config.py	Every constant the project reads
src/costing.py	The cost model against the AED 35 M ceiling
src/dataset.py	Assembles the ML training tables
src/drawings.py	Section, elevation, circulation, planting, facilities
src/figures.py	The analysis charts
src/models.py	The four models
src/plan.py	SINGLE SOURCE of the crescent geometry
src/solar.py	Sun position and shadow ray-tracing
src/viz.py	Analysis module
run_analysis.py	Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

- tools/__init__.py
- tools/build_docs.py
- tools/build_links.py
- tools/build_notebook.py
- tools/build_reports.py
- tools/build_submission_pdfs.py
- tools/organize_repo.py
- tools/report_content.py
- tools/sync_film.py
- tools/sync_portal.py
- tools/sync_submission.py

THE DATA, AS ISSUED (7)

data/raw/construction_unit_rates_aed.csv	Source dataset
data/raw/dubai_climate_normals_ncm.csv	Source dataset
data/raw/dubai_population_dsc_2023.csv	Source dataset
data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	41 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Al Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD_PARKS_MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (1)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.